Week 10 Member Report

Zeping Liao

Backend contracts, durable jobs, database integrity and recovery

Prepared for submission on 13 October 2026

Source version: CS30-1_Week10_20261013_22ccbfa899ec

Workstream outcome

Week10 adds guarded opt-in execution paths while retaining durable job identity, source permissions and publication checks. The owner partition contains 169 complete files: 163 unchanged, four changed and two added. Responsibility for this module is separate from personal executor attribution. The exact comparison baseline is final_source04_with_reports_matched5_and_token_cache_20261009_01; the Week9 archives and reports remain preserved [F].

Inherited Week9 foundation

Week9 already delivered backend support for versioned learning state, memory, notes, practice assessments and source-linked review. Durable request identity, idempotent submission, job claims, cancellation and restart recovery already existed. Saved model configurations, compatibility stages and separate answer/checker activation were also inherited.

The Week9 report records isolated backup/restore checks across 65 tables, bounded authorization controls and recovery of an original queued task without another POST. Its Linux installation and PostgreSQL integration evidence has explicit partial-gate and environment limits. These historical results are useful starting points; they are not new tests of the frozen Week10 release. Database models, migration files and scripts/release/backup.py, common.py and restore.py are unchanged in this owner partition [F].

Complete files changed after the baseline

backend/app/core/config.py adds checker schema/output/payload options, V23 query selection, the exact transport binding and the optional startup-prewarm flag. The prewarm default remains false, ordinary query preparation remains V22, and qualification transport/session/question defaults remain unset. Validators reject incomplete bindings and incompatible source-contract/checker combinations.

backend/app/modules/answering/qualification_transport.py is new. It limits ordinary_qa_compact_v2 to one configured session and SHA256 of the exact normalized API question. Eligible requests must be direct textbook QA without reading context, learner attempts or practice/tutor controls. Out-of-scope submissions fail CONFLICT. A command containing a binding without its transport also fails CONFLICT; legacy behavior applies only when both fields are absent [N].

backend/app/modules/answering/service.py freezes the schema, output, payload and transport choices into the queued command. The worker-side GenerationRequest consumes that frozen command, so subsequent Settings changes cannot retarget an accepted request. The service validates the bound schema request before generation. The transport implementation is integrated here; it is distinct from the startup work in worker.py.

backend/app/worker.py adds optional retrieval-cache preparation after normal locks and stale recovery, before the first job claim. Eligible interactive PostgreSQL workers use a separate read-only Session on the same Engine, then roll back and close it. Unsupported queues/dialects skip preparation; failure records expose only the exception class. Ordinary retrieval still validates normally after an unsuccessful preparation [W].

conversation/query_v23.py is new, with explicit dispatch added in conversation/practice_context.py and the answering service. The bounded current-statement/assignment rules preserve the complete original question and provenance coordinates; unmatched or ambiguous requests delegate to V22. October11 adoption and subsequent assignment controls are documented as software repairs, not successful live-answer qualification [Q].

Verification and actual outcomes

The October9 prewarm affected suite passed 48 tests in 1.44 seconds, with Ruff checks passing. These free controls exercised cache identity, epoch/source changes, rollback/close and startup ordering without provider requests or real database connections. The later matched native browser pair completed two full-checker citation/refresh journeys. Send-to-complete DOM was 18.7143/12.1465 seconds, while launch-to-completion was 30.2040/29.4479 seconds. Startup cost increased; one fixed-order pair cannot establish causal speedup or warm p95. The subsequent Q1/Q2 successes used different workers, leaving continuous same-worker acceptance open [W,G].

The native129 affected suite passed 202 tests in 9.67 seconds and Ruff. Independently reviewed source evidence SHA256 is f7684342815f0b32fafc81740102f7ca03948101458d13ad1d33c120e140f9a5. An authenticated API/isolated SQLite/queued-command/worker/GenerationRequest control reached the engine guard and deliberately stopped at FREE_CONTROL_STOP. It validates the exercised transport path, not delivery of a model answer [N].

Actual S02 made one checker request: HTTP200, finish_reason=stop, 8462 total tokens and 2223.083 ms API time. All seven citation groups were assessed, but SOURCE_REFERENCE_OUTSIDE_ACTUAL_CITATIONS produced TECHNICAL_STOP. technical_valid=false; semantic acceptance, correctness and false-reject remain null. No answer was published and no retry occurred. The recorded CNY0.019558 is a protective tariff upper bound, not an invoice or browser latency. S01 has no live session, submission or dispatched request [N].

Limits and Week11 plan

The freeze establishes source identity, not a new whole-suite, installation or recovery PASS. Separate release-time Mock browser QA passed within the scoped limits recorded below. Current canonical counts remain five verified, 38 pending, eight incomplete and two external; project_complete=false. Full role/object coverage, restored source visibility, live recovery, independent scientific review and formal p95 remain open [G].

First, complete finite cancellation/restart/recovery journeys on authored isolated data. Recover original request IDs, inspect model/usage receipts before any retry, and verify no duplicate paid work or hidden UNKNOWN state. Include stale bindings and later Settings changes.

Second, exercise backup/restore and migration idempotence against the frozen schema, then inspect permission, source revocation, notes/history and queued-job state after restore. Record table preservation and resource closure independently from UI success.

Third, finish a same-worker multi-question browser itinerary with startup, Send, complete DOM and failure censoring recorded separately. Review S01 origin restrictions, settlement and close-failure behavior before any future finite authorized experiment. Keep opt-ins disabled in the release and place subsequent implementation in Week11.

Evidence references

F: SOURCE_FREEZE_FINAL.json, report-inputs/owner06-final.json and the retained Week09 Zeping report.

W: docs/execution/20261009-week10-worker-cache-prewarm.md and its affected validation receipt.

Q: docs/execution/20261011-checker85-prospective-quality-results.md and 20261011-checker86-diagnosis-and-local-repairs.md.

G: docs/execution/CURRENT_TWELVE_SECTION_ENGINEERING_GAP_MATRIX_01.md.

N: docs/execution/20261013-native129-recheck-and-week10-release-handoff.md.

Final release verification — 13 October 2026

The final source freeze is CS30-1_Week10_20261013_22ccbfa899ec. A release QA correction supplies the actual transport helper to an AST-based test harness; production behavior is unchanged. The original failed run is retained. After public tokenizer/DEV fixture preparation, the final unit run passed 4,803 tests and 215 subtests, with 28 skips and 40 warnings. Foundation, contracts, chat scope and targeted Ruff passed. TypeScript, all 241 frontend tests in 27 files, and the frontend build passed.

Actual native Edge checks passed login, New chat, complete cited mock prose, exact persisted answer/history identities, citation opening, refresh and desktop/mobile layouts. All 17 exercised runtime input hashes match the final freeze and all owned services closed. This used existing dependencies, fresh SQLite, an authored corpus and mock models. Clean dependency installation, the final full PostgreSQL/corpus launcher, live quality, formal p95 and human learning effects remain unverified. Public model hydration separately passed. Evidence: release/validation/ and RELEASE_MANIFEST.json in the complete project.